Volatility Band Width: Estimating and Backtesting Volatility Bands for the S&P 500

K. Mohamed Harith · Erdős Institute Quantitative Finance Bootcamp, Fall 2026 · github.com/Harry140896/Volatility-band-width

Question. Can volatility models give reliable 95% intervals (bands) for the S&P 500's next-day return? And if so, how should volatility be estimated to make them more reliable?

Data. Daily closing prices of the S&P 500 index (Yahoo Finance, ^GSPC), converted to log returns. Models are calibrated on **2005–2017** (including the 2008 crisis) and tested out of sample on **2018–2026** (2,223 days, including the COVID-19 crash). No test-period data is used to set any parameter.

Models. All four models model the daily return as $r_t = s_t * z_t$, where $z_t \sim N(0,1)$, with band $\pm 1.96 * s_t$; they differ only in how s_t is estimated, each fixing a weakness of the previous one:

1. **Constant GBM:** one volatility for every day, the 2005–2017 standard deviation (s= 1.19% per day). It cannot adapt to calm or turbulent markets.
2. **Rolling GBM:** the standard deviation of the last m = 160 days rolling dates (chosen on 2007–2017). It adapts, but slowly, with all days weighted equally. The 161th day drops out abruptly.
3. **EWMA:** exponentially decaying weights, $s_t^2 = \lambda s_{t-1}^2 + (1-\lambda) r_{t-1}^2$, with λ= 0.94 (RiskMetrics). Recent shocks count most.
4. **GARCH(1,1):** $s_t^2 = \omega + \alpha r_{t-1}^2 + \beta s_{t-1}^2$, estimated by maximum likelihood ($\alpha = 0.103$, $\beta = 0.880$; parameters chosen using the period 2005-2017). It adds a pull towards a long-run volatility level.

Misses in the test period. Miss rates: constant GBM 4.9%, rolling GBM 7.3%, EWMA 5.8%, GARCH 5.2%. Constant GBM's misses are packed into crises (2020, 2022, April 2025); EWMA's and GARCH's are spread out.

Validation. Kupiec's test checks the *number* of misses; Christoffersen's checks whether they *cluster*.

Model	Right number? (Kupiec)	Independent? (Christoffersen)
Constant GBM	Yes (p = 0.83)	**No** (p < 0.001)
Rolling GBM	**No** (p < 0.001)	**No** (p < 0.001)
EWMA	Yes (p = 0.11)	Yes (p = 0.09)

Model	Right number? (Kupiec)	Independent? (Christoffersen)
GARCH(1,1)	Yes (p = 0.64)	Yes (p = 0.69)

Constant GBM passes Kupiec only because opposite errors cancel: 15.0% misses in 2020 (band too narrow) and 0% in 2023 (band too wide). After one of its misses, the next day is a miss 24.8% of the time, against 3.9% otherwise.

Conclusion. Constant volatility gets the average right and the extremes wrong; EWMA and GARCH, which update volatility daily from the previous day's return, deliver close to 95% coverage in both crisis and calm years, with GARCH the most reliable narrowly.

Limitations: We have assumed that all log returns are normally distributed. In fact, normal shocks understate fat tails. Moreover we have performed the tests on one market and one split. It can be extended further to incorporate more scenarios.